SIMATS ENGINEERING
ASSESSMENT TOOL 1 Experiments

COURSE NAME: Computer Networks **COURSE CODE: CSA07**

COURSE OUTCOME COVERED

CO3: *Analyze the performance of core network protocols such as TCP, UDP, HTTP, and DNS under varying conditions*
Assessment Tool Weightage: 40%
SOFTWARE: CISCO PACKET TRACER, WIRESHARK

EXPERIMENTS

QUESTIONS:4 **Total Marks:40**

1. Design a simple LAN consisting of two PCs connected through a switch using Cisco Packet Tracer. Configure appropriate IP addresses for both hosts and verify connectivity using the ping command. Simultaneously capture the packet exchange using Wireshark. Identify the Ethernet frame fields such as Source MAC Address, Destination MAC Address, Ether type , and Frame Check Sequence (FCS)**.** Explain how the Data Link layer uses MAC addressing to ensure successful frame delivery within the local network.

   AIM:
   To design and configure a simple LAN using Cisco Packet Tracer with two PCs connected through a switch, assign appropriate IP addresses, verify connectivity using the ping command, and analyze the captured Ethernet packets using Wireshark by identifying the Source MAC Address, Destination MAC Address, Ether Type, and Frame Check Sequence (FCS) fields.

   PROCEDURE
   1. Open Cisco Packet Tracer and place two PCs and one switch in the workspace.
   2. Connect both PCs to the switch using Copper Straight-Through cables.
   3. Configure IP addresses: PC1 – 192.168.1.1 /24 and PC2 – 192.168.1.2 /24.
   4. Open the Command Prompt on PC1 and use ping 192.168.1.2 to verify connectivity.
   5. Capture the ping packet exchange using Wireshark on the connected network interface.
   6. Examine the captured Ethernet frame and identify the Source MAC, Destination MAC, Ether Type, and FCS fields.
   7. Analyze how MAC addressing at the Data Link layer enables the switch to forward frames to the correct PC within the LAN.

IMPLEMENTATION:

Screenshot 1:

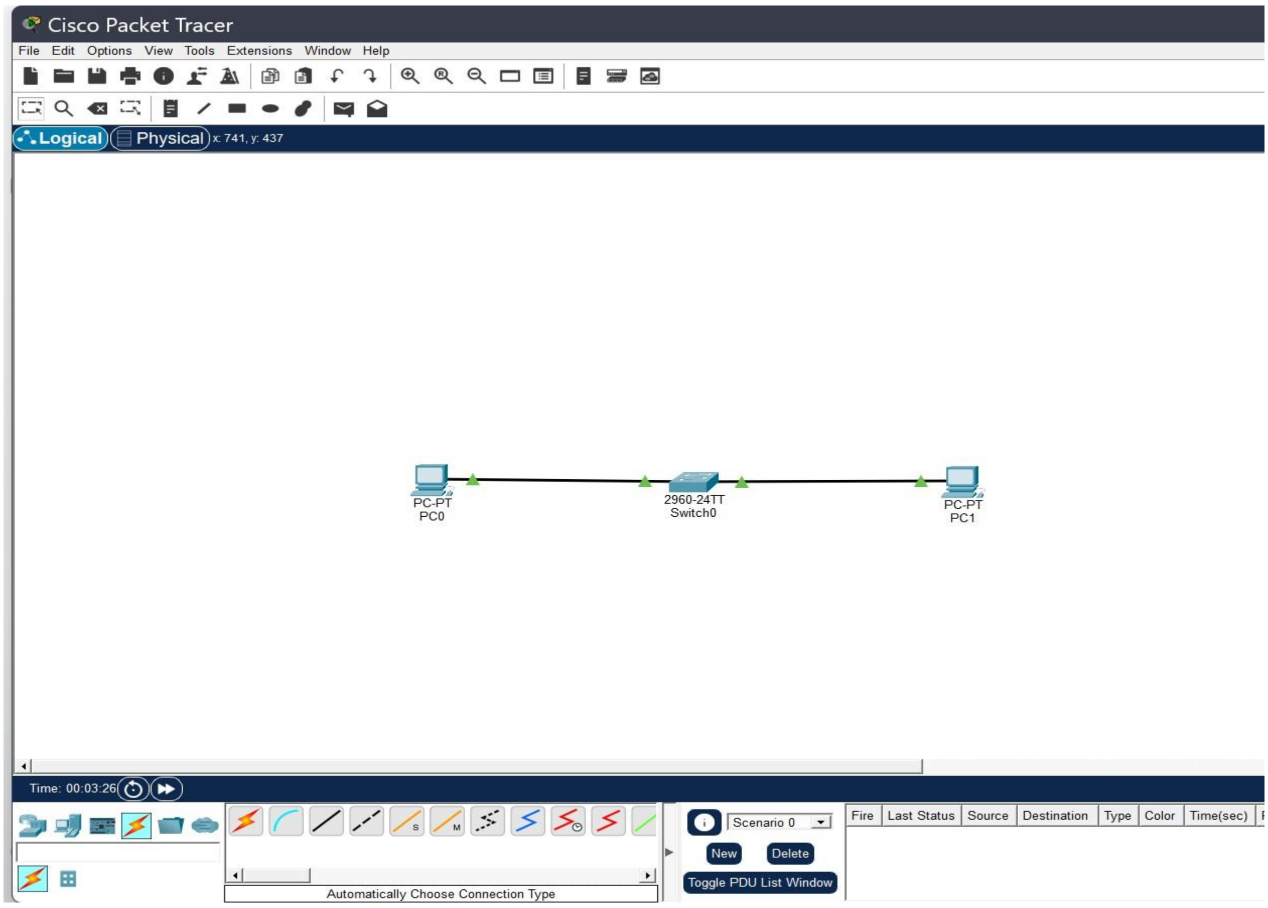


Fig.1 two pc's connected through a switch

Explanation:

Two PCs are connected to a network switch using Ethernet cables. The switch enables the PCs to communicate with each other within the same LAN by forwarding data frames based on their MAC addresses.

Screenshot 2:

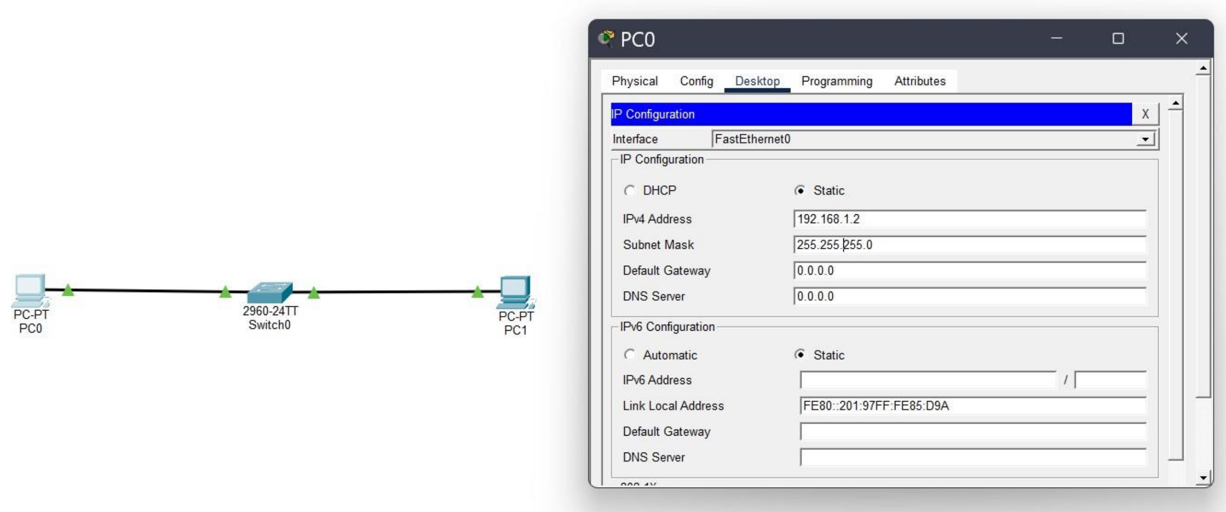
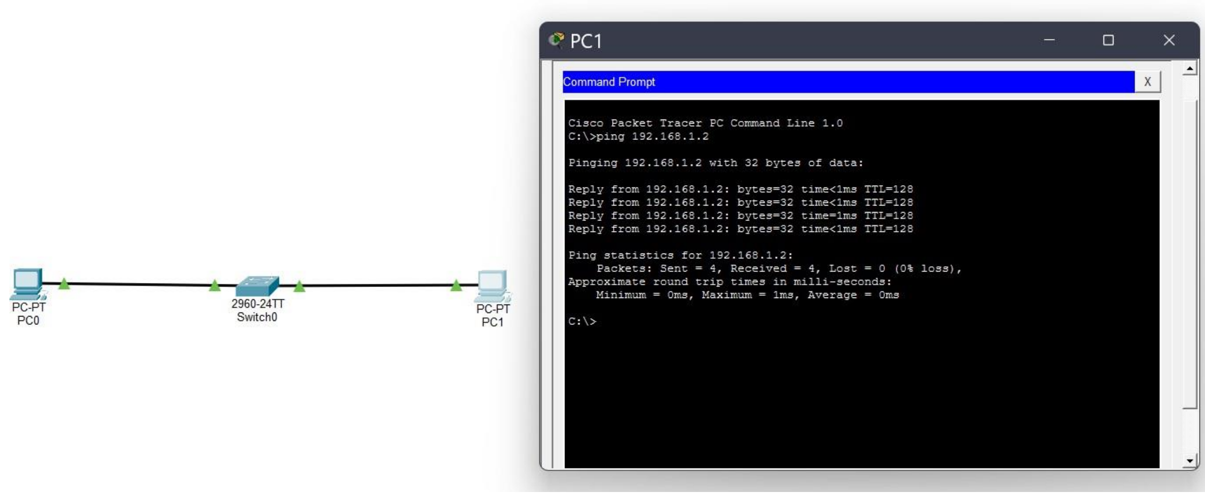


Fig2: IP configuration

Explanation:
IP addresses were assigned to both PCs in the same network. PC1 was configured with 192.168.1.1 and PC2 with 192.168.1.2, using the subnet mask 255.255.255.0. This allows both PCs to communicate within the same LAN.

Screenshot 3:

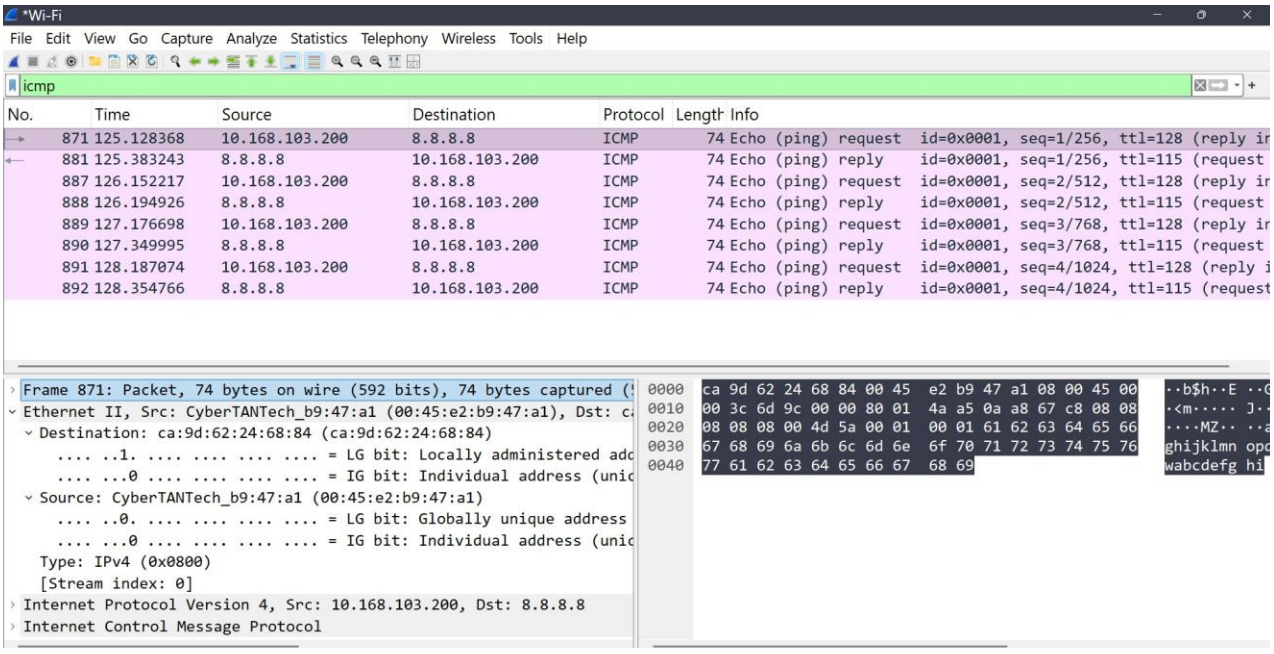
Fig 3: connectivity using ping command



Explanation:
The ping command was used to test connectivity between the two PCs.
PC1(192.168.1.1) successfully received replies from PC2 (192.168.1.2), confirming that the LAN connection was working correctly.

Screenshot 4:

Fig :4 packet exchange using Wireshark



Explanation:

Wireshark was used to

capture and the

ICMP Echo Request and

Echo Reply packets generated during the ping test. The captured packets show the communication between the source and destination devices.

The Ethernet frame fields:

ETHERNET FIELD	VALUE
Destination MAC address	Ca:9d:62:24:68:84
Source MAC Address	00:45:e2:b9:47:a1
Ether Type	0x0800 — IPv4
Protocol	ICMP

Data Link Layer Explanation:

- The Data Link Layer is responsible for reliable communication between devices
- within the same local network.
- It uses **MAC addresses** to identify the source and destination devices. When
- frame is transmitted, the **Source MAC Address** identifies the sender, while

- **Destination MAC Address** identifies the intended receiver.
- The switch examines the destination MAC address and forwards the Ethernet frame
- through the appropriate port.
- The **Ether type** identifies the network-layer protocol carried in the frame, such as
- IPv4 (0x0800).
- The **FCS** is used for detecting errors in the Ethernet frame.

Result:

- A LAN consisting of two PCs connected through a switch was successfully
- designed and configured in Cisco Packet Tracer.
- IP addresses were assigned to both PCs, and connectivity was verified successfully
- using the **ping command**.
- The Ethernet frame was analyzed using Wireshark, and the **Source MAC Address,**
- **Destination MAC Address, and Ether Type** were identified.
- The experiment demonstrated how the Data Link Layer uses MAC addressing to
- ensure successful frame delivery within a local network.

3.Construct a local area network with multiple hosts connected through a switch in Cisco Packet Tracer. Clear the ARP cache on one of the hosts and initiate communication with another host using the ping command. Capture the communication in Wireshark and identify the ARP Request and ARP Reply packets. Analyze the sender and target MAC/IP addresses and explain how the Address Resolution Protocol resolves logical IP addresses into physical MAC addresses before data transmission.

ANSWER:

Aim:

To construct a Local Area Network (LAN) with multiple hosts connected through a switch using Cisco Packet Tracer, clear the ARP cache of a host, and use the ping command to initiate communication. The ARP Request and ARP Reply packets are captured and analyzed using Wireshark to understand how ARP resolves an IP address into the corresponding MAC address before data transmission.

Procedure:

1. Open Cisco Packet Tracer, add one switch and three PCs, and connect all PCs to the switch using Copper Straight-Through cables.
2. Configure the IP addresses of the PCs in the same network, for example 192.168.1.10, 192.168.1.20, and 192.168.1.30 with subnet mask 255.255.255.0.
3. Open the Command Prompt of one PC and check its ARP table using the arp -a command. Clear the ARP cache using the appropriate ARP command.
4. Start Wireshark packet capture and initiate communication from the selected PC to another PC using the ping command.
5. Apply the Wireshark filter arp and identify the ARP Request packet, where the sender broadcasts a request to find the MAC address associated with the target IP address.
6. Identify the ARP Reply packet, where the destination PC responds with its MAC address. Note the sender and target MAC/IP addresses in both packets.
7. Observe the subsequent ICMP Echo Request and Echo Reply packets to verify that communication between the two hosts is successful.
8. Analyze the captured packets and explain how ARP maps the logical IP address to the physical MAC address before the actual data transmission takes place.

Implementation-1:

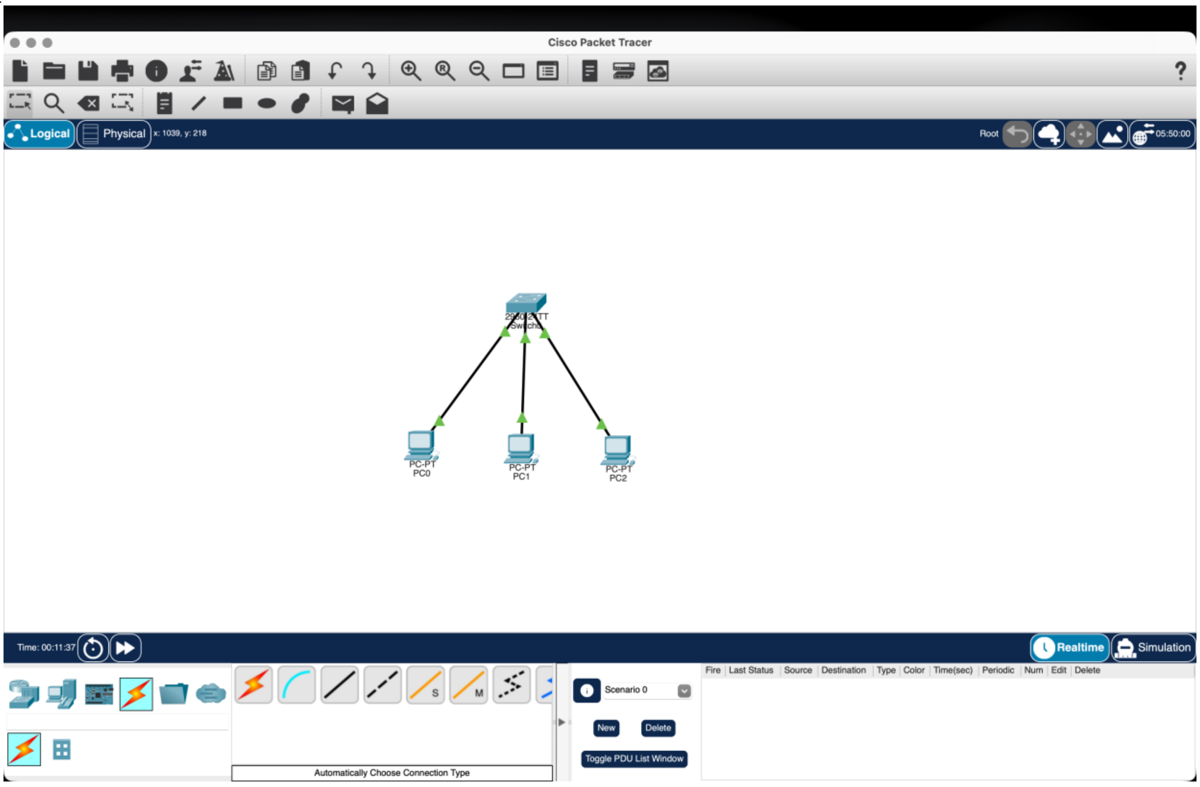


Fig1:LAN Topology with Multiple Hosts Connected Through a Switch

Explanation:

In Cisco Packet Tracer, **three PCs are connected to a single switch using Ethernet cables** to form a Local Area Network (LAN). Each PC is assigned a unique IP address within the same network, such as 192.168.1.1, 192.168.1.2, and 192.168.1.3, with the subnet mask 255.255.255.0. The switch provides communication between the connected hosts, allowing them to exchange data within the LAN. Connectivity is verified by using the ping command between the PCs.

S. No.	Device	IP Address	Subnet Mask	Purpose
1	PC0	192.168.1.1	255.255.255.0	Source host
2	PC1	192.168.1.2	255.255.255.0	Destination host
3	PC2	192.168.1.3	255.255.255.0	Additional LAN host
4	Switch	—	—	Connects all hosts

Implementation-2:

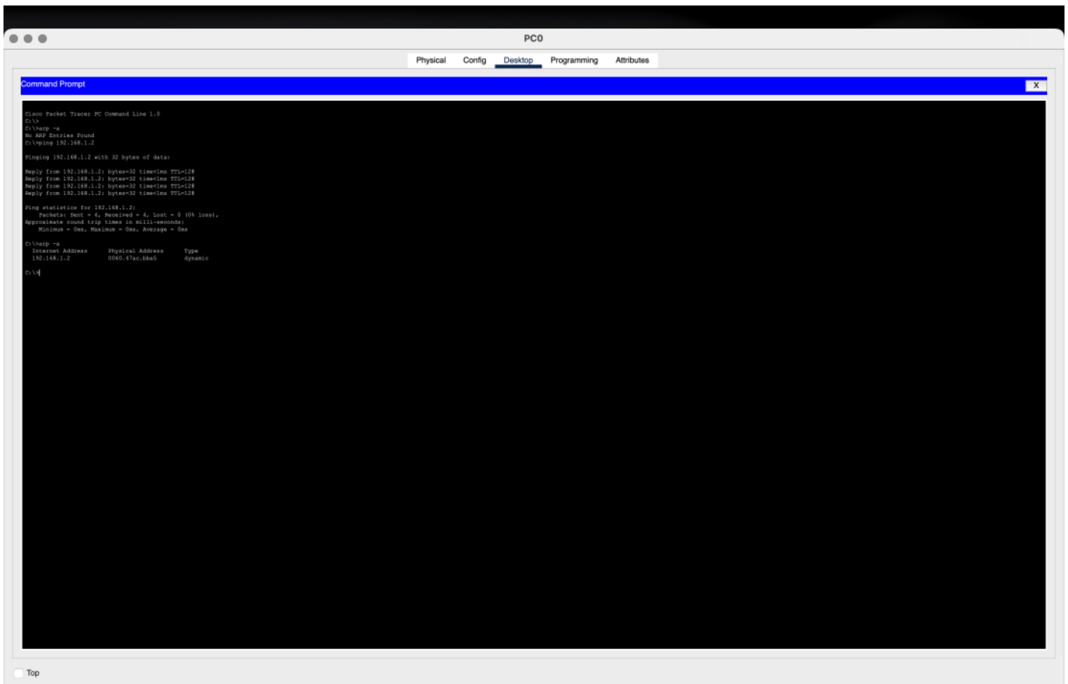


Fig2:ARPCache Verification and Ping Communication

Explanation:

The figure shows the **PC0 Command Prompt** in Cisco Packet Tracer. Initially, the ARP table is empty, as indicated by **"No ARP Entries Found."** PC0 then sends a ping to PC1 at IP address **192.168.1.2**. The four successful replies confirm that communication between the two hosts is working correctly. After the ping, the arp -a command displays the dynamically learned MAC address corresponding to 192.168.1.2. This demonstrates that ARP successfully resolved the destination IP address into its MAC address for communication.

Parameter	Result
Source IP	192.168.1.1
Destination IP	192.168.1.2
Ping Packets Sent	4
Ping Packets Received	4
Packet Loss	0%
ARP Status Before Ping	No ARP Entries Found
ARP Status After Ping	MAC address learned dynamically
Communication Status	Successful

Implementation-3:

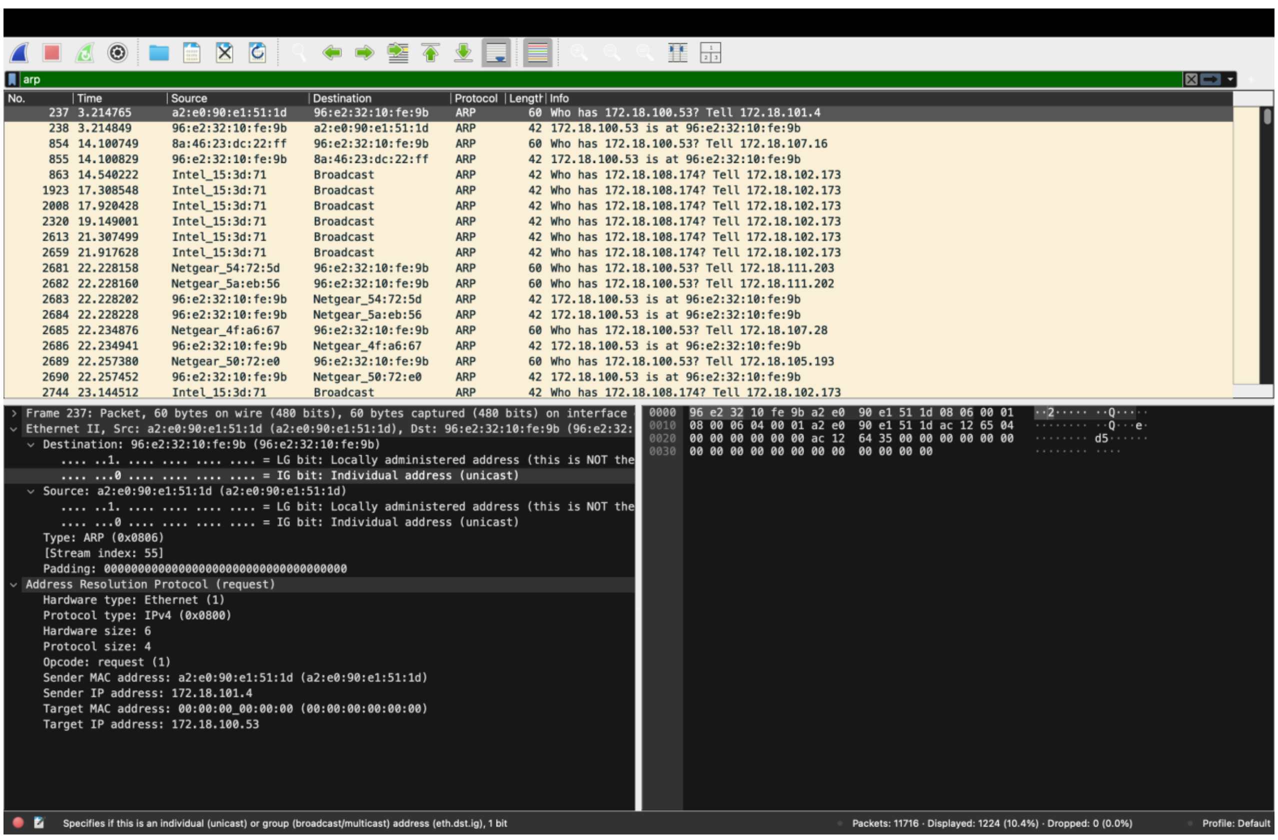


Fig3: ARP Request Packet Captured in Wireshark

Explanation:

The ARP Request is sent by the device with IP address **172.18.101.4**. Since it does not know the MAC address corresponding to **172.18.100.53**, it sends an ARP Request asking which device owns that IP address. The target MAC is shown as 00:00:00:00:00:00 because the MAC address is not yet known.

Field	Value
Packet Type	ARP Request
Sender MAC	a2:e0:90:e1:51:1d
Sender IP	172.18.101.4
Target MAC	00:00:00:00:00:00
Target IP	172.18.100.53

Implementation-4:

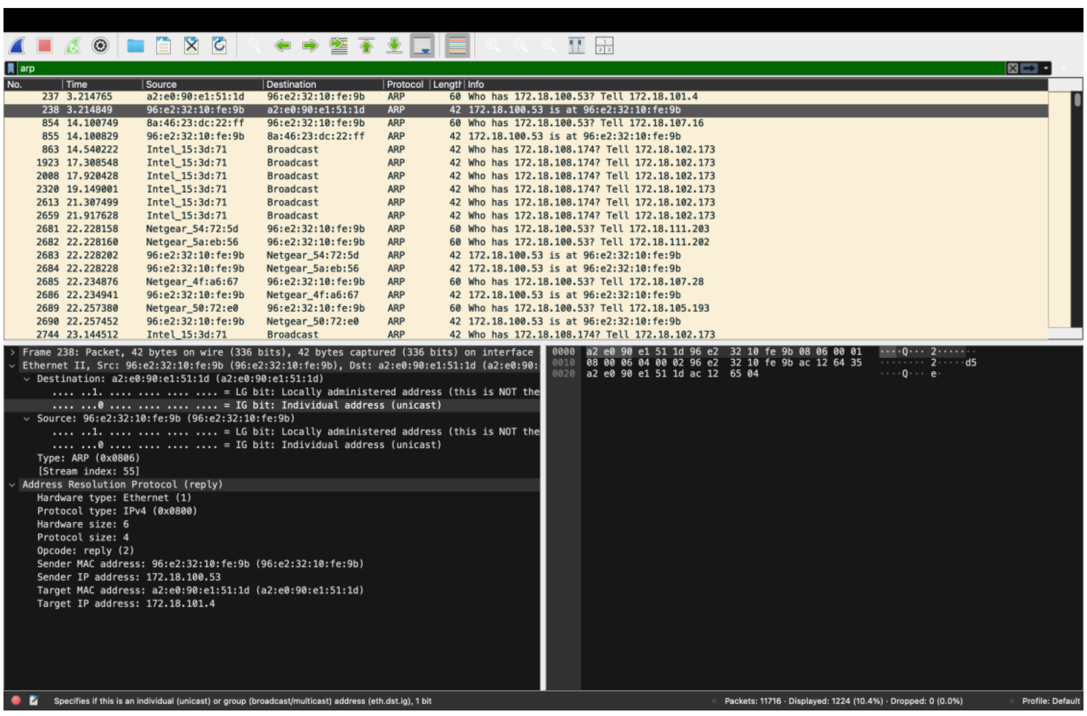


Figure 4: ARP Reply Packet Captured in Wireshark

Explanation

The ARP Reply is sent by the device with IP address **172.18.100.53** in response to the ARP Request. It provides its MAC address **96:e2:32:10:fe:9b** to the device with IP address **172.18.101.4**. This allows the requesting device to associate the destination IP address with its physical MAC address and proceed with data transmission.

Field	ARP Reply
Packet No.	238
Sender MAC	96:e2:32:10:fe:9b
Sender IP	172.18.100.53
Target MAC	a2:e0:90:e1:51:1d
Target IP	172.18.101.4

Analysis and Explanation of ARP:

From the captured Wireshark packets, the **ARP Request** shows that the sender IP address is **172.18.101.4** with MAC address **a2:e0:90:e1:51:1d**. The target IP address is **172.18.100.53**, while the target MAC address is initially unknown (00:00:00:00:00:00).

The **ARP Reply** is sent by **172.18.100.53**, whose MAC address is **96:e2:32:10:fe:9b**. It provides this MAC address to the sender **172.18.101.4**.

How ARP works:

1. The sender knows the **destination IP address**, but not its MAC address.
2. It broadcasts an **ARP Request** asking, *"Who has 172.18.100.53?"*
3. The device having that IP address responds with an **ARP Reply** containing its MAC address.
4. The sender stores the **IP-to-MAC mapping** in its ARP cache.
5. The sender can now use the destination **MAC address** to deliver the Ethernet frame.
6. Therefore, ARP resolves the **logical IP address** into the **physical MAC address** required for data transmission on the local network.